CAL HACKS 13.0 · HYPHENATE TRACK

Stochastic

Merge an add-on's books into the platform's, flag the lines a human must decide, and test the result against the lender's own limit.

THE SEAT

A portfolio controller at a private equity backed company, right after an add-on closes. The lender tests a leverage limit on the combined company, and the next deal waits until two sets of books are merged.

Today: a controller, a spreadsheet, and weeks. Compliance certificates are due 45 to 60 days after quarter end (self-reported, Aleph; terms vary per deal).

WHAT IT DOES

1. **Map.** A Mapper agent matches each target line to a platform line and tags covenant debt.
2. **Flag.** It stops on what it must not decide alone. Code computes the effect of each answer.
3. **Combine and test.** A Preparer combines the books with tools, a Reviewer checks the tie-out, a human approves.
4. **Remember.** The controller's answer becomes a policy. The next run does not ask again.

THE MOMENT: A REAL PUBLIC DEAL

Kewaunee Scientific acquires Nu Aire. Does the combined company stay under 2.50x?

PNC loan agreement, Section 4.8(a): Senior Funded Bank Debt to EBITDA below 2.50 to 1.00. Kewaunee's "financing liability" lines hold a sale-leaseback, PNC debt, and subordinated seller notes, and the agreement excludes two of them. The Mapper flags the lines; the controller decides.

THE AGREEMENT'S OWN READING

0.76x · OK

Senior Funded Bank Debt 21,633K over covenant EBITDA 28,379K. About $49.3M of covenant headroom; the next add-on still needs the $5M basket and the revolver.

READ AS FILED, EVERY FINANCING LINE

2.84x · Over

71,955K over reported EBITDA 25,312K. One reading of one line decides whether the lender's limit is breached.

TRUST

- A gateway is the only writer. Agents cannot approve or post.
- Every number comes from the Preparer's own tool runs, chained by run id; a model-made number, or an exclusion no controller made, is refused.
- A hash-chained, append-only log of every action, denials included.

STATEFUL AGENTS

- Each role keeps its Droid session across deals.
- Answers become policies the gateway writes. Second run: no flag, ready in 58 s.
- A restart loads the record, the log, and the memory back from Postgres.

BUILT WITH

- **Factory Droid SDK:** the three agents, Claude Sonnet 5.5 and Haiku 5.5.
- **Tiger Cloud:** Postgres with an append-only TimescaleDB log; each flag's what-ifs in a zero-copy fork.
- **Figma:** the console's design system.
- FastAPI, Next.js, 176 tests.

Inputs: 54 Kewaunee and 53 Nu Aire statement lines from SEC filings (10-Q, 8-K/A Ex 99.1 and 99.2, 8-K Ex 10.1). The honest limits are on the back. The output is a combined set of books and a limit check that a human approves. It is not advice.

`github.com/cayden-h/stochastic`

READ THIS BEFORE YOU TRUST A NUMBER

What is real, and what is not

REAL, AND RUNNING LIVE

- The deal, the filings, and the lender's covenant are public. Every input amount matches its SEC filing.
- The gateway, the math tools, the tie-out, the limit test, and the hash-chained log run live on screen.
- The effect of each flag answer is computed by the tools. When the Mapper leaves out the count-everything reading, the gateway adds it, labelled as its own.
- Agent memory is policies the gateway wrote from the controller's answers. It is not a change to model weights.

1. The statement lines come from public filings, not from the companies' internal books.

2. Account-level detail is not public, so we do not claim it. We map statement lines, not trial balances.

3. Agent text comes from the backend named on screen: Droid (Mapper on Claude Sonnet 5.5, Preparer and Reviewer on Haiku 5.5), or the scripted rules if Droid fails.

4. The fiscal years differ (Kewaunee April 30, Nu Aire September 30), so the combined income statement adds two different years.

5. The combined table is historical plus target. No purchase accounting is applied, so it is not the filed pro forma.

6. Two inputs to covenant EBITDA are our estimates: the Statesville cash rent we deduct ($1,970K), and transaction costs, set to zero. Each company's latest fiscal year stands in for the agreement's trailing twelve months.

7. The filing calls its own reclassification table, Note 2, preliminary. We use it only to check which line goes where.

Also not built: the fixed charge coverage test in Section 4.8(b), other deals, and other situations. No practitioner has used the tool yet.